Lecture 03.12: The range() Function

📌 Executive Summary

The range immutable sequence type: start, stop, step parameters, lazy evaluation, constant memory O(1), and index slicing.

🔬 Theoretical Foundations & Principles

1. The Immutable Sequence Type `range`

In Python 3, `range` is not a generator or list; it is an immutable sequence type representing arithmetic progressions:

$$\text{range}(\text{start}, \text{stop}, \text{step}) = \{\text{start} + i \cdot \text{step} \mid 0 \leq i < k\}$$

Where the number of elements k is:

$$k = \max\left(0, \left\lceil \frac{\text{stop} - \text{start}}{\text{step}} \right\rceil\right)$$

2. Strict $\mathcal{O}(1)$ Space Footprint

A `range` object stores only three integers: `start`, `stop`, and `step`. Whether generating 10 numbers or 10 billion numbers, its memory footprint remains exactly 48 bytes!

💻 Production Implementation & Code

```
import sys

r_small = range(10)
r_huge = range(1_000_000_000)

print("Size of range(10):", sys.getsizeof(r_small), "bytes")
```

```
print("Size of range(1 billion):", sys.getsizeof(r_huge), "bytes")
print("Range indexing r_huge[500]:", r_huge[500])
```

⚙️ Algorithmic Complexity & Systems Analysis

- **Time Complexity:** Detailed in respective algorithmic operations.
- **Auxiliary Space:** Memory bounds and CPython frame management.